DermaVision AI — Complete Project Audit, Task Roadmap & Module 5 Progress

Date & Time: October 27, 2026
Project: DermaVision AI: Skin Pigmentation Detection, Analysis & Future Skin Health Prediction
Scope: Verification of Modules 1–4, Task Roadmap Definition, and Full Implementation of Module 5 (Part A: GAN, Part B: VAE, Part C: Generative AI)

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1. Executive Summary

This document provides a comprehensive record of the conversation, thorough audit of completed modules (1 to 4), curriculum task definition for Module 5, and full implementation records for all three components of **Module 5: Generative Intelligence**:
* **Module 5 Part A: Generative Adversarial Networks (DCGAN & cDCGAN)**
* **Module 5 Part B: Variational Autoencoders (ConvVAE) & Case-Based Reasoning Retrieval**
* **Module 5 Part C: Generative AI Multimodal Synthesis, Clinical Reports & Skincare Chatbot**

DERMAVISION AI MODULE STATUS

Module	Core Curriculum Topics	Status	Health / Tests
Module 1	ANN, MLP, ReLU, Softmax, Adam	VERIFIED	100% Passed
Module 2	Backprop, Regularization, He/Xavier Init	VERIFIED	100% Passed
Module 3	CNN, Transfer Learning, YOLO, U-Net	VERIFIED	Weights Intact
Module 4	RNN, LSTM, GRU, BiLSTM Temporal Tracking	VERIFIED	100% Passed (6)
Module 5A	DCGAN, cDCGAN Synthetic Skin Generation	COMPLETED	100% Passed (8)
Module 5B	ConvVAE & Clinical Similarity Retrieval	COMPLETED	100% Passed (8)
Module 5C	Generative AI Skincare Consultation Engine	COMPLETED	100% Passed (8)

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2. In-Depth Audit of Modules 1 to 4 (Teammates' Contributions)

Module 1: Artificial Neural Network (ANN/MLP) for Clinical Risk

- **Curriculum Mapping**: Use an ANN/MLP to analyze structured user information (age, gender, skin type, UV exposure, lifestyle, skincare routine) and predict pigmentation risk. Use ReLU in hidden layers, Softmax/Sigmoid for output, and Adam optimizer for training.
- **Audit & Code Verification**:
- **Architecture**: `src/model.py` implements `DeepSkinDiseaseANN` with a 78-dimensional feature input, 3 hidden layers (256 $\to$ 128 $\to$ 64) with `ReLU`, `Dropout(0.3, 0.25, 0.2)`, and 6-class

`Softmax` output.

- **Preprocessing**: `src/preprocessing.py` standardizes 4 numerical features (`age`, `fitspatrick`, `diameter_1`, `diameter_2`) and one-hot encodes 74 categorical/symptom dummy variables.
- **Score Engine**: `src/score_engine.py` calculates:
  1. Skin Health Score (0–100)
  2. Pigmentation Risk Score (0–100)
  3. Clinical Severity Score (0–100)
- **Verification**: Executed `tests/test_pipeline.py` $\to$ **100% Passed** with 93.19% confidence on test samples.
- **Documentation**: `docs/module1_documentation.md`.

Module 2: Optimization, Regularization & Weight Initialization

- **Curriculum Mapping**: Train the ANN/CNN using Backpropagation. Apply Dropout and L2 Regularization to reduce overfitting. Use He/Xavier Initialization for faster convergence.
- **Audit & Code Verification**:
- **Experiments**: Documented in `DermaVision_Module2_Experiment_Results.csv` across 10 distinct configurations comparing:
  - Weight Initializations: Xavier/Glorot vs. He/Kaiming vs. Default PyTorch.
  - Optimizers: Adam vs. RMSprop vs. SGD.
  - Learning Rates: 0.01, 0.001, 0.0001.
  - Regularization: Dropout (0.2–0.3) and L2 weight decay (10^{-4}).
- **Top Performing Model**: Xavier Initialization + Adam ($lr=0.001$) achieved **80.22% test accuracy** and **0.7092 Macro F1**.
- **Artifacts**: Best checkpoint `DermaVision_Module2_Final_DeepANN.pth` with confusion matrix and loss curves in `results_for_Weight_Initialization_Optimization/`.

Module 3: Computer Vision (CNN, Transfer Learning, YOLO, U-Net)

- **Curriculum Mapping**: CNN classifies skin conditions (Melasma, Hyperpigmentation, Acne Marks, Freckles, Healthy Skin). Transfer Learning (ResNet50/EfficientNet/MobileNetV2) improves accuracy. YOLO detects and localizes pigmentation patches, acne, dark spots, or moles. U-Net segments the affected skin region and calculates the percentage of pigmentation.
- **Audit & Code Verification**:
- **Transfer Learning Classifiers**:
  - `resnet50_best.pt` (94.4 MB)
  - `transfer_learning_runs/efficientnet_b0_best.pt` (16.4 MB)
  - `transfer_learning_runs/mobilenet_v2_best.pt` (9.2 MB)

- **YOLO Localization**: Trained YOLO detector in `yolo_runs/detection_pigmentation/weights/best.pt` and `yolo26n.pt` with PR and F1 curves.
- **U-Net Semantic Segmentation**: Checkpoint `unet_pigmentation_best.pt` (93.3 MB) with skin masking via YCrCb color space.
- **Quantification Reports**: `pigmentation_report.csv` and `pigmentation_class_summary.csv`.
- **Documentation**: `Details_of_module3_Subhankar.md`.

Module 4: Temporal Progression & Clinical NLP (RNN, LSTM, GRU, BiLSTM)

- **Curriculum Mapping**: Store skin images over time and use LSTM/GRU to predict whether pigmentation is improving or worsening. BiLSTM analyzes dermatologist notes or user skincare journals to understand treatment progress and feedback.
- **Audit & Code Verification**:
- **Recurrent Progression Architectures**: `src/module4_models.py` implements `SkinProgressionLSTM` and `SkinProgressionGRU` with multi-horizon projection heads (+1 month, +3 months, +6 months) and a 3-class trajectory classifier (`Improving`, `Stable`, `Worsening`).
- **Clinical Feedback NLP**: `ClinicalJournalBiLSTM` classifies dermatologist notes and patient journals into `Positive`, `Neutral`, or `Adverse` reactions.
- **Data Engine**: `src/module4_sequence_dataset.py` formulates 4-visit temporal sequences tracking health score, pigmentation %, severity, risk, diameters, and adherence over time.
- **Checkpoints**: `models/module4_lstm_progression_best.pth`, `models/module4_gru_progression_best.pth`, `models/module4_bilstm_notes_best.pth`, and `models/module4_vocab.json`.
- **Verification**: Cleaned duplicate filename to standard `tests/test_module4.py` $\to$ **All 6 tests passed**.
- **Documentation**: `Details_of_module4_MouliAnkita.md`.

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3. Module 5 Implementation Records (Complete)

Part A: Generative Adversarial Networks (GAN)

- **Curriculum Requirement**: *"GAN generates synthetic skin images to increase the training dataset."*
- **Architecture**: `src/module5_gan.py` implements `SkinLesionGenerator`, `SkinLesionDiscriminator`, `ConditionalSkinGenerator` (conditioned on 6 clinical classes: ACK, BCC, MEL, NEV, SCC, SEK), and custom weight initialization.
- **Training & Results**: 25 epochs trained on Apple MPS GPU (`train_module5_gan.py`), logged in `DermaVision_Module5_GAN_Results.csv`, generating 120 synthetic skin lesion images organized into class folders (`synthetic_skin_dataset/`).
- **Artifacts**: `models/module5_gan_generator_best.pth`, `Module5_GAN_Synthetic_Samples.png`, `Module5_GAN_Loss_Curves.png`.

- **Documentation & Notebook**: `Details_of_module5_GAN.md`, `Module5_GAN_Synthetic_Augmentation.ipynb`.

Part B: Variational Autoencoder (VAE) & Case Similarity Search

- **Curriculum Requirement**: *"VAE learns compact representations of skin images for similarity search and visualization."*
- **Architecture**: `src/module5_vae.py` implements `SkinLesionVAE` ($z \in \mathbb{R}^{32}$ with Reparameterization Trick and β-VAE loss) and `SkinCaseSimilarityEngine` (Case-Based Reasoning via Cosine Similarity and Euclidean Distance).
- **Reference Gallery**: Indexed 150 clinical cases with confirmed histological outcomes in `models/module5_vae_case_gallery.pt`.
- **Visualizations**: `Module5_VAE_Reconstruction_Grid.png`, `Module5_VAE_Latent_Space_tSNE.png` (2D clustering), `Module5_VAE_Latent_Interpolation.png` (continuous morphological traversal).
- **Documentation & Notebook**: `Details_of_module5_VAE.md`, `Module5_VAE_Similarity_and_Latent_Space.ipynb`.

Part C: Generative AI Multimodal Consultation Engine

- **Curriculum Requirement**: *"Generative AI creates personalized skincare reports, explains results in simple language, answers skincare questions, and suggests routines based on the analysis."*
- **Multimodal Synthesizer**: `src/module5_genai.py` fuses:
- Module 1/2: Health Score, Severity Score, Pigmentation Risk, Diagnostic Probabilities.
- Module 3: U-Net Pigmentation Percentage (%) and YOLO detection.
- Module 4: 1M, 3M, 6M LSTM progression trajectory and BiLSTM note sentiment.
- Module 5 (VAE): Top-3 historically matched cases and anomaly atypicality score.
- **Personalized Clinical & Skincare Reports**: `generate_clinical_and_skincare_report()` provides:
  1. Executive Summary & Plain Language Diagnosis Translation.
  2. 1M, 3M, 6M Trajectory & Prognosis Narrative.
  3. Evidence-Based Historical Case Comparison.
  4. Step-by-Step AM Protocol (Defense, Niacinamide/Vitamin C, Mineral SPF 50+).
  5. Step-by-Step PM Protocol (Cellular Repair, Barrier Fortification, Ceramides, Azelaic Acid).
  6. Targeted Actives vs. Strictly Contraindicated Ingredients.
- **Interactive AI Skincare Chatbot**: `DermaVisionConsultationBot` answers patient inquiries regarding sun safety, ingredient interactions, flare-ups, and long-term forecasts.
- **Web API Endpoints** (in `app.py`):
- `POST /api/module5/generate-report`: Complete report generator endpoint.
- `POST /api/module5/chat`: Context-aware conversational AI endpoint.
- `POST /api/module5/similar-cases`: VAE latent retrieval endpoint.
- `POST /api/module4/forecast`: LSTM progression trajectory endpoint.

- **Documentation**: `Details_of_module5_GenAI.md`.

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4. Full Test Suite Execution Record (30 Tests Passing)

Command:

```
python3 -m unittest discover -s tests -p "test_*.py"
```

Output:

```
[Test Setup] Running Module 4 test suite on: CPU
   [PASS] Longitudinal dataset generation & feature shapes verified.
   [PASS] SkinProgressionLSTM multi-horizon forward pass verified.
   [PASS] SkinProgressionGRU forward pass verified.
   [PASS] ClinicalJournalBiLSTM text logits and softmax verified.
   [PASS] End-to-end multi-horizon forecast data verified.
   [PASS] Clinical notes NLP inference function verified.

[Module 5 Test Setup] Running GAN test suite on: MPS
   [PASS] SkinLesionGenerator output shape and [-1, 1] bounds verified.
   [PASS] SkinLesionDiscriminator output shape and [0, 1] probabilities verified.
   [PASS] ConditionalSkinGenerator class conditioning verified.
   [PASS] ConditionalSkinDiscriminator forward pass verified.
   [PASS] DCGAN custom weight initialization verified.
   [PASS] Synthetic skin sampling and [0, 1] normalization verified.
   [PASS] Synthetic image grid export verified.
   [PASS] GAN Dataset Augmentation Engine successfully verified.
   [PASS] Multimodal patient context synthesis verified.
   [PASS] Clinical & skincare report structure verified.
   [PASS] AM/PM structured regimens verified.
   [PASS] Plain language explanations & knowledge base verified.
   [PASS] Chatbot sunscreen consultation response verified.
   [PASS] Chatbot ingredient guidance & safety warnings verified.
   [PASS] Chatbot longitudinal trajectory response verified.
   [PASS] Stateless chat API helper verified.

[Module 5 VAE Test Setup] Running VAE test suite on: MPS
   [PASS] VAE Encoder output shapes (mu, logvar) verified.
   [PASS] VAE Decoder reconstruction shape and [0, 1] range verified.
   [PASS] Reparameterization trick verified (deterministic eval, stochastic train).
   [PASS] Beta-VAE loss function verified.
   [PASS] Clinical case gallery indexed successfully in latent space.
   [PASS] Case similarity search and top-K ranking verified.
   [PASS] Lesion anomaly score & reconstruction error verified.
   [PASS] Gallery serialization (save & load) verified.

Ran 30 tests in 0.938s — OK (100% Passed)
```

Pipeline Integration Test:

```
python3 tests/test_pipeline.py
# PIPELINE TEST PASSED SUCCESSFULLY!
```

5. Summary of Project Deliverables

Module	Core Deliverables	Verification Status
Module 1	`DeepSkinDiseaseANN`, Preprocessing, Score Engine	Verified & Passing
Module 2	Weight Init Ablation (Xavier vs. He), Loss Curves	Verified & Passing
Module 3	ResNet50, EfficientNet, YOLO Detection, U-Net Segmentation	Checkpoints Intact
Module 4	`SkinProgressionLSTM`, `ClinicalJournalBiLSTM`, Sequence Dataset	6 Tests Passing
Module 5A (GAN)	`SkinLesionGenerator`, `ConditionalSkinGenerator`, Augmentation	8 Tests Passing
Module 5B (VAE)	`SkinLesionVAE`, Case Similarity Engine, t-SNE Projections	8 Tests Passing
Module 5C (GenAI)	Multimodal Context Fusion, Clinical Report Engine, Chatbot, Flask APIs	8 Tests Passing

6. How to Test the Whole System (Step-by-Step Guide)

You can test the entire multimodal system using any of the following methods:

Method 1: All-In-One Master Integration Test (Recommended)

Run the automated end-to-end master test that exercises every neural network and Flask endpoint:

```
python3 test_full_system.py
```

This script validates:
- Module 1/2 DeepANN feedforward inference and score engine calculations.
- Module 3 checkpoint integrity and quantification CSV reports.
- Module 4 LSTM multi-horizon trajectory forecasting (+1m, +3m, +6m).
- Module 5A DCGAN image sampling and normalization.
- Module 5B VAE latent gallery indexing and similarity retrieval.
- Module 5C Generative AI personalized skincare report generation and live chatbot query.

Method 2: Comprehensive Unit Test Discovery Suite

To run all 30 modular unit tests across all test suites simultaneously:

```
python3 -m unittest discover -s tests -p "test_*.py"
```

Individual module tests can also be run independently:

```
# Module 4 (Recurrent Progression & NLP):
python3 tests/test_module4.py

# Module 5A (GAN Synthetic Augmentation):
python3 tests/test_module5_gan.py

# Module 5B (VAE & Case Similarity Search):
python3 tests/test_module5_vae.py

# Module 5C (Generative AI Skincare Consultation):
python3 tests/test_module5_genai.py

# Module 1 Pipeline Integration:
python3 tests/test_pipeline.py
```

Method 3: Live Interactive Web Dashboard Test

To test the visual user experience in your web browser:
1. Start the Flask server:
`bash python3 app.py`
2. Open your browser and navigate to:
`http://localhost:5000`
3. Click any **Quick Preset** button (e.g. *High-Risk Melanoma*, *Basal Cell Carcinoma*, *Benign Mole*, or *Actinic Keratosis*).
4. Click **Execute DermaVision AI Prediction**.
5. Observe the unified multimodal results:
- **Diagnostic & Severity Cards** (Modules 1 & 2)
- **6-Class Softmax Probability Distribution** (Modules 1 & 2)
- **Module 4 Recurrent Trajectory & +1M/+3M/+6M Projections**
- **Module 5B VAE Top Similar Cases** retrieved from latent space
- **Module 5C Tailored AM/PM Skincare Protocol & Active Ingredients**
- **Interactive AI Skincare Chatbot**: click any prompt chip or type questions like:
- *"Which sunscreen should I use and how often?"*
- *"Which active ingredients are safe for my skin?"*
- *"What does my 3-month forecast mean for me?"*

Method 4: Visual Jupyter Notebook Inspection

For visual demonstrations and theoretical charts, open any of the interactive notebooks in VS Code or Jupyter:
* `Module5_GAN_Synthetic_Augmentation.ipynb`

* `Module5_VAE_Similarity_and_Latent_Space.ipynb`
* `Module4_Longitudinal_Skin_Progression.ipynb`
* `Module2_Integration.ipynb`